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Efficient DNA Isolation method from fresh tissue of banana leaf without using liquid nitrogen and β-mercaptoethanol

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Abstract- In the realm of advanced molecular genetic analysis, techniques such as phylogenetic analysis, marker-assisted selection, purity analysis, gene mapping, and DNA fingerprinting, heterotic analysis play a vital role. However, one of the most critical and cumbersome steps in these processes is the high-throughput DNA isolation process. This step is not only labor-intensive but also time-consuming, often serving as a bottleneck in molecular genetic analysis. This study introduces a simplified, cost-effective modified CTAB-DNA isolation method that can be efficiently executed in a laboratory with basic equipment. Notably, this approach eliminates the need of liquid nitrogen and hazardous chemical like β-mercaptoethanol, thereby enhancing laboratory safety and cost-effectiveness. The amount and quality of isolated Deoxyribonucleic acid (DNA) depends on starch and non-cellulosic occurrence in the leaf wall of *Musa paradisiaca* (Banana). However, the described method overcomes these challenges, yielding high-quality DNA well-suited for PCR analysis. This modified CTAB-DNA isolation approach offers a rapid, easy, inexpensive, and reliable solution for researchers working on banana, making it an attractive option for molecular genetic studies.

Key words: DNA isolation, Banana leaf, Liquid nitrogen and β-mercaptoethanol

INTRODUCTION

Musa paradisiaca (banana) is 2nd leading fruit crop of India, surpassed only by *Mengifera indica* in terms of importance and cultivation.[1] Bananas are the most widely consumed fruit across all socio-economic classes in India, attributed to their year-round availability, affordability, diverse varieties, appealing taste, high nutritional value, and medicinal properties, making them a staple fruit for the masses.[2] Bananas offer immense export and foreign exchange earnings, driven by their widespread cultivation across over 120 countries globally. India stands out as the world's largest banana producer, accounting for 14.2 million tonnes of the global output of 86 million tonnes. Other significant banana-producing countries include the Philippines, Indonesia, Brazil, Ecuador, China, Mexico, Thailand, and Colombia. With a dominant share of 27% in global banana production, India solidifies its position as a leader in the industry. The utilization of modern farm equipment enables farmers to provide optimal growing conditions, unlocking the full varietal potential of the crop and facilitating financially sustainable cultivation practices.

The Bihar region exhibits a strong preference for locally adapted banana varieties, such as Batisa and Chiniya, due to their distinctive taste and finger

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characteristics. These varieties have high market value among. In recent years, molecular biology developed as a powerful tool for analyzing genetic diversity, offering valuable insights into the genetic makeup of these locally adapted varieties. By employing techniques such as DNA fingerprinting and genetic marker analysis, researchers can better understand the genetic relationships between different banana varieties, ultimately informing breeding programs and conservation efforts[3] or genetic fidelity of in vitro grown plantlets[4]. Molecular techniques yield highly reliable results, unaffected by environmental factors like soil quality, temperature & humidity. Such consistency enables researchers to accurately assess genetic diversity, identify specific genetic markers, and establish precise relationships between different banana varieties. The reliability of molecular techniques makes them an indispensable tool for plant breeding, genetic research, and conservation efforts, ultimately contributing to the development of improved banana cultivars.[5] Random Amplified Polymorphic DNA (RAPD) has been widely employed molecular markers and has proven successful in detecting genetic similarities and differences in tissue-cultured materials and cultivated plants. This enables the initiation of informed breeding programs.[6] The current research study focused to tests the fidelity standard of micropropagated plantlets developed through tissue culture of banana cultivars Battisa and Chiniya. However, ensuring the genetic stability of tissue-cultured plantlets is equally crucial, and understanding genetic stability or variation is essential for quality control in large-scale banana propagation. This information is vital for establishing a reliable micropropagation system that produces genetically identical, stable, and disease-free plants, which is a critical consideration before commercial release. This study seeks to address this key issue, providing valuable insights into the genetic fidelity of micropropagated banana plantlets.

MATERIALS & METHODS

Plant materials

Banana suckers of two varieties under current studies were procured from local farmer's field.

Table 1- Index of varieties used in the current study

Sl. No.	Varieties
1	Batisa
2	Chiniya

Genomic DNA extraction

Two distinct banana cultivars (detailed in Table 1) were utilized for genomic DNA extraction. To initiate the process, a 0.3 g sample of fresh, unopened pale yellow leaf tissue was crushed in a mortar and pestle until a uniform pale yellowish color was achieved. This was done in the presence of CTAB extraction buffer, comprising 0.2 percent (w/v) Cetyl Hrimethylmmonium bromide, 0.022M EDTA, and 1.4M Sodium chloride. Subsequently, a new 2 ml centrifuge tube was prepared, and the homogenized pale yellowish tissue sample was added, accompanied by the extraction buffer, to facilitate the DNA extraction process. Following incubation, the homogenized tissue sample was centrifuged for two minutes at 4000 revolution per minute. Know 1 ml homogenized sample was taken in a fresh 2 ml centrifuge tube, and 400 microliter of isoamyl alcohol:chloroform (1:24) was added. The mixture was thoroughly combined until the color became faint, and then centrifuged at 4000 revolution per minute for ten minutes. The overlying aqueous part was taken in a fresh 2 ml centrifuge tube, and the isoamyl alcohol:chloroform step was performed again until the overlying aqueous part set off transparent. Once solution set off transparent, the overlying part was carefully taken in a fresh 2 ml centrifuge tube through a pipette. Next, 500 microliters of chilled isopropyl alcohol were carefully added along the inner wall of the centrifuge tube, followed by gentle inversion of the mixture. The addition of iso-propyl alcohol caused the DNA to precipitate, forming a visible bubble-like structure.

The resulting DNA was carefully taken in a fresh 2 ml tubes employing wide-bore pipette tips. The aqueous state was slowly dispensed without discarding the DNA (deoxyribonucleic acid) cluster by tilting the centrifuge tube. The cluster DNA was then washed with 500 µl of 76% ethyl alcohol consisting of 0.2M Sodium acetate, then through 76% ethyl alcohol consisting 10mM Ammonium acetate.

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Then after the isolated DNA was air-dried by upending the tube on filter paper until decapitation of smell

of ethyl alcohol. The isolated DNA was then dissolved in 100 microliter of 1X TE (10 mM Tris-HCl, 1 mM EDTA) buffer, & 5 microliter RNAse solution (0.01 gm/ml) was added. The resulting DNA solution was incubated at 37 degree Celsius for 1 hour to decompose any remaining RNA impurities. Ultimately, the dissolved DNA (deoxyribonucleic acid) was stored at -4 degree Celsius until further work.

DNA purification & quantification

The concentration (ng/µl) of isolated DNA was quantified using a double beam uv-vis spectrophotometer (LABTRONICS MODEL LT-2900), measuring the absorbance at 260 nm & 280 nm. The purity of the DNA was examined by the ratio of absorbance at 260 nm to 280 nm using a using a double beam uv-vis spectrophotometer (LABTRONICS MODEL LT-2900). Agarose gel electrophoresis was performed to estimate integrity and quality of the isolated DNA. 7 µl isolated DNA was mixed with 1 µl of 10X loading dye and loaded onto a 0.8% (w/v) agarose gel containing 1 µg/ml ethidium bromide in 1X TAE buffer. Electrophoresis was performed for 1 hour at 60 V.

After electrophoresis, the gel was visualized and photographed employing gel doc system, allowing for the examination of DNA quality and integrity based on the presence and intensity of DNA band.

Amplification of DNA

Polymerase chain reaction was carry out in a thermal cycler (Prima-96^{PLUS}, Hi-media) employing a 20 µl reaction mixture consisting of 12.8 microliter without nuclease water, 2.0 microliter 10X green taq buffer (Thermo scientific), 2.0 microliter 2mM dNTPs (Thermo scientific), 2.0 microliter of random primers having concentration 5.0 pmol/ µl, 0.2 µl 5.0 U/ µl Dream Taq Green DNA Polymerase and 1.0 µl extracted genomic DNA.

The thermal cycling program consisted of initial denaturation for 3 minutes at 94 °C, followed by 30 cycles of denaturation at 94 °C, annealing temperature 40 °C for 1 minute, extension at 72 °C for 1 minute and, at the end of 30 cycle reaction, a final extension at 72 °C for 7 minutes was set. The authenticity & applicability of this method were validated employing 2 random primers, OPN-09 and OPN-10 (Table 2), of genomic DNA amplification.

Table 2- Details of random primers employed to validate the amplification by PCR employing genomic DNA isolated from two varieties of *Musa paradisiaca*.

SN	Primer	Primer Sequence 5'-3'	Annealing temperature (°C)
I.	OPA-1	CAGGCCCTTC	40
II.	OPA-2	TGCCGAGCTG	40

RESULT & DISCUSSION

Among various techniques, the CTAB (Cetyltrimethylammonium Bromide) method proved to be the most effective for isolating high-quality plant genomic DNA. The DNA isolation protocol consist of 3 main steps: sample grinding, extraction using isoamyl alcohol & chloroform, and DNA precipitation. The selection of genomic DNA isolation protocol depends on many factors, consisting of the genomic DNA size and quantity, desired purity for downstream applications, and considerations of cost and time involved. The DNA isolation method standardized is a slight modification of the technique suggested by Suman *et al.* (2020)[7]. This modified CTAB DNA extraction method does not require liquid nitrogen and this method eliminating hazardous chemical i.e. β-mercaptoethanol.[8] The extracted DNA quality was examined using the ratio of absorbance at 260/280 nm. Batisa & Chiniya have absorbance ratios of 1.884 and 1.818 at 260 and 280 nm, respectively.

The isolated DNA exhibited high quality, with 260/280 nm absorbance ratios ranging from 1.8 to 2 (Table 3). Furthermore, the amplified DNA products displayed strong intensity on agarose gel (Fig. 1), demonstrating successful compatibility with random primers OPA-1 and OPA-2.

Table 3: Spectrophotometer measurements of two banana genomic DNA concentrations

Sl. No	Variety Name	A260	A280	A260/A280 Ratio	DNA Concentration (µg/µl)
I.	Batisa	0.049	0.026	1.884	0.488
II.	Chiniya	0.040	0.022	1.818	0.498

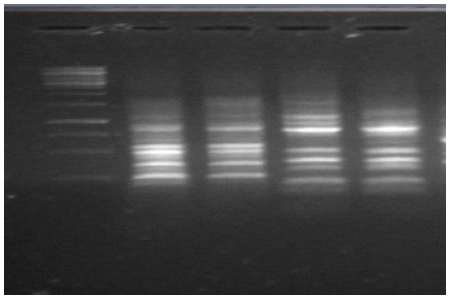


Fig 1: Amplification pattern of genomic DNA using employing OPA-1 & OPA-2

CONCLUSION

In conclusion, without need of liquid nitrogen and β-mercaptoethanol, this extraction protocol for banana DNA extraction is reliable and consistent method that performs well for DNA extraction procedure. It may be utilized for molecular genetic analysis, techniques such as phylogenetic analysis, marker-assisted selection, purity analysis, gene mapping, and DNA fingerprinting, heterotic analysis.

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